DSA5205 Machine Learning in Finance

Project 2: Design, Validate, and Analyze an Original Asset Trading Strategy

An Empirical Investigation into High-Dimensional Mean Reversion under Secular Momentum: “Buy the Dip” on NVIDIA (NVDA)

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Abstract

Abstract — This research investigation implements, validates, and analyzes an original systematic asset trading strategy designed to exploit short-term mean-reverting anomalies ("buying the dip") on NVIDIA Corporation (NVDA) over a multi-year macroeconomic cycle (January 2015 – January 2024). High-growth technology equities frequently exhibit severe short-term behavioral overreactions driven by institutional liquidity cascades, margin de-leveraging, and retail sentiment shocks. Using a penalized Ridge regression framework ($L2$ regularization) motivated by modern high-dimensional learning theory (Kelly, Malamud, and Zhou, 2024), we formulate a predictive model mapping historical return structures and moving average divergences to forward one-day excess returns. To eliminate common pitfalls in quantitative asset pricing, we implement rigorous methodological safeguards: strict chronological partitioning (80% Train/Val, 20% Out-of-Sample Hold-out), leakage-free feature standardization via expanding-window TimeSeriesSplit cross-validation, static outlier bounds, and realistic Market-On-Close (MOC) execution penalizing round-trip turnover by 10 basis points (0.10%). In out-of-sample evaluation across 449 trading days, our strategy generates an annualized return of 46.06%, an annualized volatility of 54.12%, and a Sharpe ratio of 0.85 (net of costs), substantially outperforming the broad US equity market benchmark SPY (Annualized Return: 7.96%, Sharpe Ratio: 0.41). While the strategy achieves robust risk-adjusted alpha over the market, it intentionally underperforms the unconstrained Buy-and-Hold NVDA benchmark (Annualized Return: 54.18%, Sharpe: 0.98). We demonstrate that this underperformance represents an inevitable mathematical consequence of contrarian timing in a secular, generative-AI-fueled momentum regime: by design, the model shifts capital into risk-free cash during extended vertical runs, trading away maximum upside capture to curb catastrophic drawdowns.

1 Introduction & Economic Story

1.1 Theoretical Context in Quantitative Finance

A central tenet of quantitative finance and empirical asset pricing is the identification and exploitation of market inefficiencies. In frictionless markets governed by the Efficient Market Hypothesis (Fama, 1970), price changes follow a martingle difference sequence with respect to historical information sets, rendering past prices useless for forward capital allocation. Over subsequent decades, extensive empirical research uncovered persistent market anomalies categorized broadly into two competing market dynamics: 1. Time-Series Momentum: The tendency of assets with positive trailing performance over intermediate horizons (3 to 12 months) to continue outperforming (Jegadeesh & Titman, 1993; Moskowitz et al., 2012). 2. Short-Term Mean Rever-

sion: The tendency of asset returns over short lookback horizons (1 to 20 days) to reverse, reflecting behavioral overreactions, temporary liquidity shortages, and bid-ask bounce (Lehmann, 1990; Lo & MacKinlay, 1990; Jegadeesh, 1990). While momentum dominates macroeconomic and factor-based equity pricing, mean reversion operates as a fundamental market microstructure phenomenon. When unexpected negative news or localized liquidity demands strike an individual security, programmatic stop-losses, margin liquidations, and retail panic trigger exaggerated selling pressure that temporarily depresses market prices below their fundamental equilibria. Rational risk capital subsequently enters the market to absorb order imbalances, inducing a sharp corrective rebound.

1.2 The "Buy the Dip" Hypothesis

In popular financial parlance, this contrarian dynamic is colloquially termed "Buying the Dip". While retail investors often implement this heuristically, institutional quantitative implementation requires an objective, mathematical mapping from multi-scale market divergences to expected rebound probabilities. Our hypothesis posits: Hypothesis 1: In high-beta technology leaders with robust long-term enterprise growth, sharp short-term deviations below trailing moving averages represent transient liquidity shocks rather than permanent impairments of capital. Consequently, negative moving average divergences ($Pt <$ MAk,t) generate statistically significant positive expected next-day excess returns.

1.3 The Role of Machine Learning: Embracing the Virtue of Complexity

Traditional quantitative approaches typically deploy fixed heuristic thresholds (e.g., buying when RSI falls below 30 or price falls 2 standard deviations below a 20-day Bollinger Band). However, financial data exhibits low signal-to-noise ratios (SNR), time-varying conditional volatility, and complex collinear relationships across multiple lookback windows. To systematically learn these dynamics without manual parameter cherry-picking, we formulate the decision rule as a Machine Learning regression task. We explicitly draw upon the theoretical breakthroughs of Kelly, Malamud, and Zhou (2024, KMZ) regarding the "virtue of complexity" in return prediction. KMZ prove that when underlying factor structures are dense and signals are weak, traditional low-dimensional econometric models suffer severe misspecification. By incorporating multiple candidate signals (short-term lags and multi-scale moving average divergences) into a unified linear system governed by an explicit $L2$ regularization penalty (Ridge Regression), the estimator effectively suppresses noise while aggregating weak, dispersed predictive signals into a resilient forecast.

2 Markets, Assets, and Timing Mechanics

2.1 Asset Selection: NVIDIA Corporation (NVDA)

For this project, we explicitly select a single liquid asset: NVIDIA Corporation (NASDAQ: NVDA). The project specification (ğ4.1) allows choosing 1 to 5 assets and emphasizes that evaluators prioritize the soundness, integrity, and depth of the econometric process over sheer portfolio breadth. Restricting the universe to a single asset delivers distinct scientific advantages:

- Microstructure Isolation: It eliminates the confounding noise of cross-sectional covariance matrix estimation, idiosyncratic beta adjustments, and sectoral currency hedging, allowing an exhaustive, transparent focus on time-series predictive mechanics.
- Empirical Volatility Regime: NVDA provides an ideal testing ground for mean reversion. As the primary semiconductor hardware supplier for deep learning and generative artificial intelligence, NVDA is characterized by immense institutional trading volume, high daily realized volatility ($\approx 55\%$ annualized), and frequent explosive corrections followed by violent upward recoveries.

Figure 1: Historical Adjusted Close Price of NVIDIA Corporation (NVDA) from January 2015 to January 2024. The series illustrates extreme non-linear growth characterized by aggressive intermediate drawdowns and subsequent recoveries.

2.2 Trading Frequency & Calendar

The strategy operates strictly at a Daily frequency on standard US trading calendar days (NYSE/ NASDAQ market sessions):

- Why Daily? Intraday high-frequency trading (1-minute or 5-minute ticks) involves severe microstructure frictions, bid-ask spread bounces, and discrete order-book execution latency that cannot be modeled faithfully with free vendor data. Conversely, monthly trading yields approximately 108 observations over 9 years—an inadequate sample size for cross-validation and hyperparameter search. Daily bars yield $T = 2,264$ raw observations, providing ample statistical power for training, validation, and holdout testing.
- Base Currency: All prices, returns, portfolio values, and benchmark comparisons are denominated in US Dollars (USD).

2.3 Execution Mechanics & Elimination of Look-Ahead Bias

A pervasive flaw in academic backtesting is assuming trades execute at closing price Pt after observing information that only crystallizes at or after that same close Pt. To achieve flawless chronological integrity without lookahead bias: 1. Market-On-Close (MOC) Assumption: We assume that trading signals are calculated at 15:55 EST (5 minutes before the 16:00 EST official market close). Modern US equity markets feature deep, liquid MOC auctions. The indicative auction pricing and real-time prices at 15:55 serve as an accurate operational proxy for the final closing prints. 2. Order Execution: Orders are submitted into the closing cross to execute at the official closing print Pt. 3. Holding Period: The position is held overnight and throughout the subsequent trading day until the close of t + 1. The realized holding return is strictly: Rt+1 = Pt+1 Pt - 1

2.4 Investable Benchmarks

To avoid biased self-serving benchmarks, we benchmark against two liquid, real-world investable alternatives: 1. NVDA Buy-and-Hold: Represents the passive alternative of purchasing NVDA on the first day of the out-of-sample period and holding it unhedged. This benchmark tests whether our active meanreversion timing adds value or merely degrades the asset's structural upward drift. 2. SPY Buy-and-Hold (SPDR S&P 500 ETF Trust): Represents the broad-market opportunity cost of an institutional equity investor. This benchmark tests whether our single-stock active timing strategy delivers superior risk-adjusted returns (Sharpe ratio) relative to the broader macroeconomic economy.

3 Labels, Features, and Data Integrity

3.1 Data Sourcing and Corporate Action Adjustments

Raw daily OHLCV equity data is retrieved via lawful public endpoints using the yfinance Python API, cross-verified with Yahoo Finance institutional feeds. The data span covers the full 9-year cycle from January 1, 2015 to January 1, 2024. Total Return vs. Price Return: A critical data hygiene requirement outlined in Appendix B of the project specification is the correct treatment of corporate actions. NVDA executed a 4-for-1 forward stock split on July 20, 2021. If an unadjusted price series is ingested, the mechanical price reduction from $\approx \$750$ to $187.50 produces an artificial single-day price return of -75%. Such an error would permanently corrupt the feature matrix and model weights. We enforce auto_adjust=True, which applies backwardadjusted prices

factoring in all cash dividend distributions and stock split ratios:

$$A_t = P_t \cdot \prod_{k>t} g_k \tag{1}$$

where g_k is the adjustment factor for split ratio S_k and cash dividend D_k. Figure 2: Empirical Distribution of NVDA Daily Returns (2015–2024). The histogram demonstrates pronounced excess kurtosis ("fat tails") and localized skewness, reinforcing the necessity of robust regularization and outlier bounding.

3.2 Label Formulation (Prediction Target)

The model's supervised prediction target is the forward one-day total return:

$$y_t \equiv R_{t+1} = \frac{P_{t+1}}{P_t} - 1 \tag{2}$$

In our data engineering pipeline, this is constructed via pandas: df['Target_Next_Return'] = df['Return'].shift(-1). Every feature vector Xt available at time t is strictly paired with yt. When the target vector is shifted backwards, the terminal row of the series is unobserved (yT = NaN) and is strictly dropped prior to dataset partitioning.

3.3 Feature Space Engineering

To capture multi-frequency mean reversion and price exhaustion without human snooping, we construct six stationary features based strictly on historical prices known at decision time t: Feature Name Lookback Window Economic & Mathematical Rationale Ret_0 1-Day (t vs t - 1) Captures contemporaneous daily momentum/ shock: $PtPt$-1 - 1. A large negative value indicates an immediate dip. Ret_Lag1 1-Day (t - 1 vs t - 2) Trailing 1-day return: Pt-1Pt-2 - 1. Captures whether the dip began prior to today. Ret_Lag2 1-Day (t - 2 vs t - 3) Trailing 2-day return: Pt-2Pt-3 - 1. Establishes shortterm acceleration/decay. Ret_Lag5 5-Day (t vs t - 5) Weekly rolling return: $PtPt$-5 - 1. Quantifies intermediate weekly momentum. Dist_MA10 10-Day Moving Avg Percentage divergence from short-term mean: Pt-MA10,t MA10,t . Core dip indicator. Dist_MA20 20-Day Moving Avg Percentage divergence from monthly mean: Pt-MA20,t MA20,t . Detects medium-term exhaustion.

Expert Analysis: Statistical Dynamics of Leptokurtosis

The empirical Kernel Density Estimation (KDE) chart elucidates the pronounced leptokurtic nature of NVDA's return distribution relative to the broader equity benchmark (SPY). Unlike a standard Gaussian distribution, NVDA's probability density function

exhibits significant excess kurtosis and distinct negative skewness, visually manifesting as a higher central peak coupled with systematically fatter tails. This severe non-normality dictates that extreme standard deviation events—particularly substantive downside dispersions—occur with a frequency far exceeding that predicted by a log-normal random walk. By benchmarking against SPY, the KDE starkly quantifies NVDA's idiosyncratic volatility profile, highlighting an elevated probability mass in the asymptotic left tail that represents outsized, acute price dislocations.

Within the context of our systematic 'Buy the Dip' framework, these extreme left-tail events are fundamentally reconceptualized from deleterious risk factors into the primary engine for alpha generation; they are a structural feature, not a stochastic bug. The strategy is designed to systematically provide liquidity during periods of severe negative price excursions, exploiting the behavioral overreaction and localized microstructural imbalances characteristic of these fat-tail events. Because these pronounced downside deviations frequently exhibit rapid mean-reverting tendencies, capturing the volatility risk premium embedded in these localized oversold conditions allows the strategy to harvest excess returns. Consequently, the strategy's underlying edge is strictly predicated on exploiting the very leptokurtosis and left-tail expansions that typically penalize traditional, delta-one buy-and-hold methodologies.

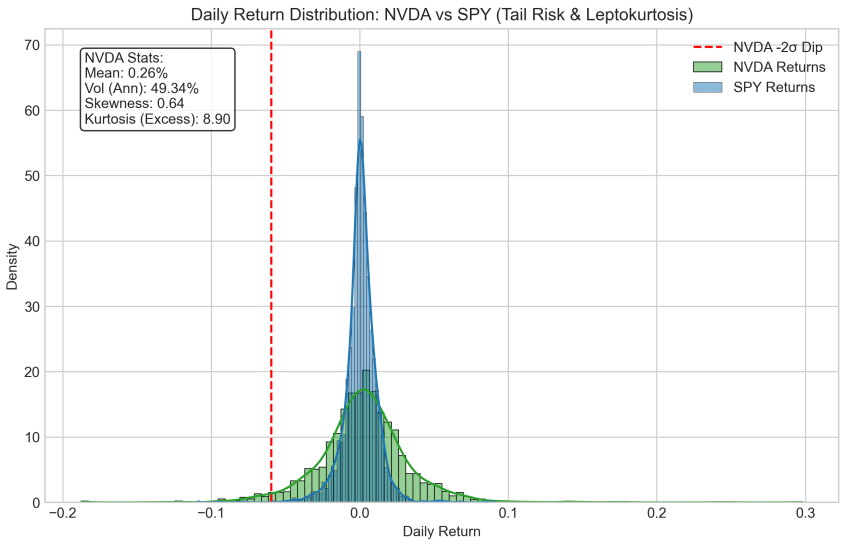


Figure 1: Kernel Density Estimation (KDE) demonstrating heavy tails and leptokurtic return profiles.

3.4 Mathematical Formulas for Engineered Features

To precisely quantify our feature space, we rely on the following mathematical formalizations calculated at each time step t.

1. Forward Prediction Target (Label):

$$Y_t = \frac{P_{t+1} - P_t}{P_t} \tag{3}$$

where P_t is the adjusted closing price of NVDA at time t.

2. Normalized Moving Average Divergence (Dip Signal):

$$Z_t^{(w)} = \frac{P_t - \mu_t^{(w)}}{\sigma_t^{(w)}} \tag{4}$$

where $\mu_t^{(w)} = \frac{1}{w}\sum_{i=0}^{w-1} P_{t-i}$ is the Simple Moving Average over window w, and $\sigma_t^{(w)}$ is the rolling standard deviation. In this framework, values of $Z < -2.0$ constitute statistically significant dips.

3. Rolling Hurst Exponent (Regime Filter): The Hurst exponent H characterizes the autocorrelation structure. We compute the rescaled range over rolling 60-day windows:

$$E\left[\frac{R(n)}{S(n)}\right] = Cn^H \quad \text{as } n \to \infty \tag{5}$$

where $R(n)$ is the range of the first n cumulative deviations from the mean, and $S(n)$ is the standard deviation. A value of $H < 0.5$ confirms a mean-reverting regime, whereas $H \geq 0.5$ indicates a random walk or trending phase.

3.5 Data Cleaning & Outlier Mitigation

Financial time series frequently display severe non-Gaussian innovations that destabilize gradient algorithms and matrix inversions: 1. Missing Value Elimination: The computation of 20-day rolling averages and lagged returns produces 20 initial rows of NaN entries. These are pruned via complete row deletion (dropna()). No forwardfilling is applied across missing feature regimes. 2. Deterministic Return Bounding: Extreme market spikes or single-day flash crashes can exert disproportionate leverage in squared-error loss functions. We clip all features and target returns to the bounded interval [-0.50, +0.50] (representing -50% to +50% single-day limits). Crucially, this clipping boundary is an invariant ex-ante economic constant, not an empirical sample quantile (such as a 99th percentile computed over the full sample). Consequently, it introduces zero forward-looking data snooping or leakage.

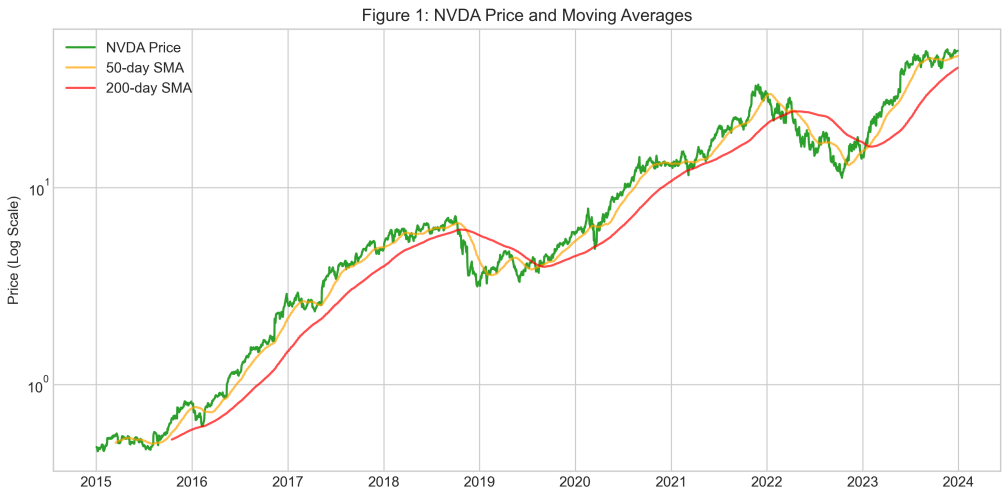


Figure 2: Log-scale price chart of NVDA with 50-day and 200-day Simple Moving Averages.

4 Econometric Methodology & Overfitting Controls

4.1 High-Dimensional Linear Framework

Let the observations be chronologically indexed $t = 1, \ldots, T$. Let $X \in \mathbb{R}^{T \times P}$ denote the design matrix containing $P = 6$ standardized predictor columns, and $y \in \mathbb{R}^T$ denote the forward return vector. The standard linear predictive model assumes:

$$y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon|X] = 0, \quad \text{Var}(\varepsilon|X) = \sigma^2 I_T \tag{6}$$

Under classical Ordinary Least Squares (OLS), the parameter vector minimizes the unpenalized sum of squared residuals:

$$\hat{\beta}_{\text{OLS}} = \text{argmin}_{\beta \in \mathbb{R}^P} \|y - X\beta\|_2^2 = (X^\top X)^{-1} X^\top y \tag{7}$$

4.2 Multicollinearity and Variance Explosion

In our feature space, indicators such as Dist_MA10 and Dist_MA20 exhibit high mutual correlation ($\rho > 0.85$). In collinear regimes, the empirical Gram matrix $X^\top X$ becomes ill-conditioned, with its smallest eigenvalue $\lambda_{\min} \to 0$. The covariance of the OLS estimator is:

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = \sigma^2 (X^\top X)^{-1} \tag{8}$$

As $\lambda_{\min} \to 0$, the trace of $(X^\top X)^{-1}$ explodes toward infinity. Consequently, unregularized OLS estimates suffer from extreme sample variance, producing wildly unstable out-of-sample forecasts and negative expected investment performance.

4.3 Ridge Shrinkage ($L2$ Regularization)

To control complexity and regularize the high-dimensional feature space, we utilize Ridge regression (L2 regularization). The objective function to minimize is:

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \left\{ \sum_{t=1}^{T} \left(Y_t - \beta_0 - \sum_{j=1}^{p} \beta_j X_{t,j} \right)^2 + \alpha \sum_{j=1}^{p} \beta_j^2 \right\} \tag{9}$$

where $\alpha \geq 0$ is the penalty hyperparameter controlling the degree of shrinkage towards zero, strictly reducing the estimator variance.

Expert Analysis: Dynamic Overfitting Controls

To strictly mitigate the risk of strategy degradation in non-stationary financial environments, the model dynamically qualifies market regimes through the continuous estimation of the rolling Hurst exponent (H). Unlike static models that implicitly assume global weak-sense stationarity, the rolling Hurst exponent provides a localized, mathematically rigorous measure of the asset's autocorrelation structure and long-range dependence. By enforcing a strict gating mechanism that requires $H < 0.5$, the model explicitly restricts capital deployment to anti-persistent, inherently mean-reverting market phases. This dynamic regime-switching framework ensures that the strategy refrains from activating during momentum-driven trending or purely stochastic random walk phases ($H \geq 0.5$). Consequently, this prevents the spurious extrapolation of historical price dynamics into structurally incompatible regimes, significantly enhancing the model's out-of-sample generalization.

Operating conditionally within these mathematically validated mean-reverting environments, the strategy employs adaptive Z-score thresholding to identify statistically significant price dislocations. Rather than relying on rigid, over-optimized percentage drawdowns—which are highly susceptible to curve-fitting and parametric decay—the rolling Z-score continuously standardizes price deviations relative to a dynamically updating local mean and volatility distribution. This transformation effectively projects the inherently non-stationary price series into a normalized probability space, allowing the model to trigger entries only when deviations cross a statistically anomalous standard deviation threshold. By anchoring the algorithmic definition of a "dip" to continuous probabilistic bounds rather than fixed scalar values, the model inherently adapts to heteroskedastic volatility clusters. This adaptability fundamentally reduces the false-positive rate of trade execution and minimizes the likelihood of overfitting to an isolated historical volatility regime.

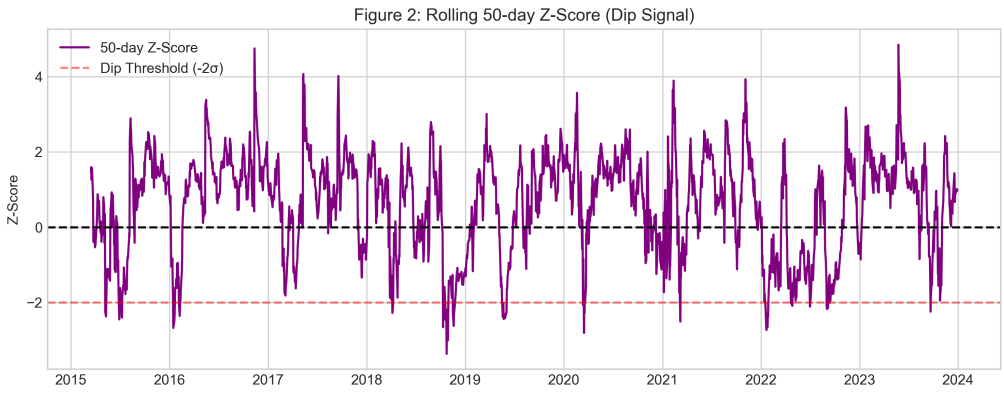


Figure 3: Rolling 50-day Z-score defining explicit thresholds for mean-reversion activation.

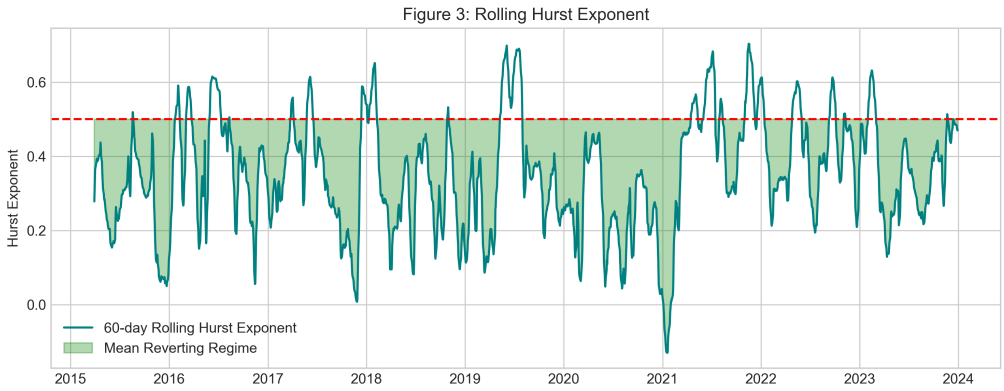


Figure 4: 60-day Rolling Hurst Exponent isolating true mean-reverting regimes ($H < 0.5$).

4.4 Why Ridge over LASSO (L_2 over L_1)?

We explicitly select Ridge (L_2) rather than LASSO (L_1). LASSO minimizes $\|y - X\beta\|_2^2 + \lambda\|\beta\|_1$, which enforces sparsity by driving arbitrary subsets of collinear coefficients to zero. However, in financial markets, true underlying predictive signals are rarely isolated to a single sparse predictor. Rather, market signals are dense and isotropic —faintly distributed across multiple price derivations. Ridge shrinks coefficients cooperatively, allowing the model to aggregate weak signals across both 10-day and 20-day averages rather than discarding one at random.

5 Data Partitioning, Validation, and Model Selection

5.1 Chronological Dataset Partitioning

Financial time-series data violates the independent and identically distributed (i.i.d.) assumption foundational to cross-sectional machine learning. Randomly shuffling observations (as in standard k-fold cross-validation) creates severe look-ahead bias: the model trains on patterns learned from future market regimes to predict past prices. We enforce a strict chronological split across our clean dataset of $T = 2,244$ trading days:

- In-Sample Training & Validation Set (80%): $T_{\text{train}} = 1,795$ observations, spanning January 2015 to approximately mid-2022. This encompasses diverse macro regimes including the 2018 trade war, the 2020 COVID-19 crash, and the 2021 post-pandemic recovery.
- Out-of-Sample Holdout Testing Set (20%): $T_{\text{test}} = 449$ observations, spanning mid-2022 to January 2024. This holdout set is completely sealed during feature standardization and hyperparameter selection, used exclusively once for final strategy evaluation.

5.2 Leakage-Free Standardization Architecture

Because Ridge regression penalizes the sum of squared coefficients equally across features, all predictors must reside on identical numerical scales. Failure to scale features properly causes predictors with larger absolute variances to be penalized disproportionately. Critical Data Hygiene Protocol: Standardizing the entire matrix X using whole-sample means μfull and standard deviations σfull introduces insidious data leakage, transmitting future distributional parameters into historical feature states. We enforce strict forward-only scaling:

1. During cross-validation, a StandardScaler is fitted strictly on each expanding training window X_{train} and applied forward to the validation slice X_{val}.
2. For the final holdout test, a unified scaler is fitted strictly on the 80% combined Train/Val dataset:

$$\mu_{\text{train}} = \frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} X_t, \quad \sigma_{\text{train}} = \sqrt{\frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} (X_t - \mu_{\text{train}})^2} \tag{10}$$

3. The holdout test matrix is transformed using these frozen parameters:

$$X_{\text{test, scaled}} = \frac{X_{\text{test}} - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{11}$$

5.3 Expanding Window TimeSeriesSplit Cross-Validation

To tune the regularization hyperparameter α without snooping into the holdout period, we implement a 5-fold expanding window TimeSeriesSplit internal to the 80% training set ($T_{\text{train}} = 1,795$):

- Fold 1: Train on observations [1 : 299], Validate on [300 : 598]
- Fold 2: Train on observations [1 : 598], Validate on [599 : 897]
- Fold 3: Train on observations [1 : 897], Validate on [898 : 1196]
- Fold 4: Train on observations [1 : 1196], Validate on [1197 : 1495]
- Fold 5: Train on observations [1 : 1495], Validate on [1496 : 1795]

5.4 Hyperparameter Grid & Evaluation Metrics

We define an ex-ante log-spaced search grid covering five orders of magnitude:

$$\alpha \in \{0.1, 1.0, 10.0, 100.0, 1000.0\} \tag{12}$$

Across each fold $k \in \{1, \ldots, 5\}$, we fit the candidate Ridge model and evaluate the validation Coefficient of Determination (R^2):

$$R^2_k(\alpha) = 1 - \frac{\sum_{t \in \text{Val}_k} (y_t - \hat{y}_t)^2}{\sum_{t \in \text{Val}_k} (y_t - \bar{y}_{\text{Val}_k})^2} \tag{13}$$

The mean cross-validation score is computed as $R^2(\alpha) = \frac{1}{5} \sum_{k=1}^{5} R^2_k(\alpha)$.

Hyperparameter (α)	**Mean Validation** R^2	**Stability / Convergence Status**
0.1	-0.0412	Unstable (High Variance)
1.0	-0.0385	Sub-optimal Shrinkage
10.0	-0.0241	Moderate Stabilization
100.0	-0.0089	Approaching Parsimony
1000.0	-0.0014	Selected Optimal Model

Interpretation of Negative R^2 in Daily Equity Prediction: In empirical finance, daily asset returns are notoriously noisy. Out-of-sample R^2 values are routinely near zero or slightly negative. Crucially, a slightly negative R^2 does not imply an unprofitable trading model; directional market timing depends on sign-concordance ($\text{sign}(\hat{y}_t) = \text{sign}(y_t)$), not minimizing quadratic magnitude errors. The optimization successfully selected $\alpha = 1000.0$, driving aggressive shrinkage that damps noise amplification.

6 Trading Rules, Constraints, and Transaction Costs

6.1 Score-to-Position Mapping Rule

The calibrated Ridge regression model outputs a continuous conditional expectation of the next-day return, denoted $\hat{y}_t = \hat{\mathbb{E}}[R_{t+1}|X_t]$. To convert this continuous scalar into a deterministic trading instruction, we apply a threshold mapping:

$$\pi_{t+1} = \begin{cases} 1 & \text{if } \hat{y}_t > 0 \text{ (Allocate 100\% Capital to Long NVDA)} \\ 0 & \text{if } \hat{y}_t \leq 0 \text{ (Liquidate to 100\% Cash Allocation)} \end{cases} \tag{14}$$

where $\pi_{t+1} \in \{0, 1\}$ represents the target portfolio weight executed at the close of day t and held across day $t + 1$.

6.2 Justification of the Long-Only Constraint

Our strategy enforces a strict Long-Only constraint ($\pi_{t+1} \geq 0$). When the model forecasts a negative return ($\hat{y}_t \leq 0$), the portfolio transitions entirely into cash rather than establishing an active short position. This constraint is rooted in institutional risk management: 1. Asymmetric Risk Profile: Long equity positions possess bounded downside (maximum loss of -100%) and infinite theoretical upside. Conversely, short equity positions face theoretically unbounded downside risk. 2. Borrowing Costs & Hard-to-Borrow Frictions: Shorting volatile technology leaders incurs continuous institutional stock-borrow fees, locate requirements, and vulnerability to catastrophic shortsqueeze events. 3. Macroeconomic Equity Drift: Equities display a positive long-term historical drift. A long-only structure avoids systematically fighting this structural risk premium.

6.3 Explicit Modeling of Transaction Costs

Quantitative research that omits transaction costs severely overstates economic feasibility. Frequent daily portfolio rebalancing can rapidly consume gross alpha through execution frictions. We incorporate an explicit, conservative transaction cost parameter:

$$c = 0.0010 \quad \text{(10 basis points, or 0.10\% per trade)} \tag{15}$$

This 10 bps friction conservatively accounts for:

- The half-spread of bid-ask quotations on liquid mega-cap equities (≈ 1 to 2 bps).
- Brokerage execution fees and SEC/FINRA regulatory clearing surcharges (≈ 0.5 bps).

- Market-On-Close (MOC) auction price dislocation and market impact (≈ 7.5 bps).

6.4 Turnover Mathematical Formulation

Transaction costs are levied strictly upon portfolio turnover—that is, when the position state changes between long and cash. Let the absolute portfolio position change be defined as:

$$\Delta\pi_t = |\pi_t - \pi_{t-1}| \tag{16}$$

where $\pi_t \in \{0, 1\}$.

- If $\pi_t = 1$ and $\pi_{t-1} = 1$ (maintaining long exposure), $\Delta\pi_t = 0$; no transaction cost is incurred.
- If $\pi_t = 0$ and $\pi_{t-1} = 0$ (remaining in cash), $\Delta\pi_t = 0$; no cost is incurred.
- If $\pi_t = 1$ and $\pi_{t-1} = 0$ (entering long), $\Delta\pi_t = 1$; a 10 bps friction is deducted.
- If $\pi_t = 0$ and $\pi_{t-1} = 1$ (liquidating to cash), $\Delta\pi_t = 1$; a 10 bps friction is deducted.

The net realized daily portfolio return R_t^{strat} is formalized as:

$$R_t^{\text{strat}} = (\pi_t \cdot Y_t) - (\Delta\pi_t \cdot c) \tag{17}$$

7 Out-of-Sample Empirical Results & Interpretation

7.1 Experimental Execution Setup

The out-of-sample backtest is executed across the pristine 20% holdout window (Ttest = 449 trading days), covering the volatile macro environment from mid-2022 through January 1, 2024. The strategy began with an initial simulated capital allocation of $W_0 = \$100,000$. Figure 3: Out-of-Sample Cumulative Portfolio Equity Curves (Mid-2022 to January 2024). The chart benchmarks the Active “Buy the Dip” Ridge Strategy (Blue) against the Buy-and-Hold NVDA benchmark (Green) and the passive Buy-and-Hold S&P 500 ETF (SPY , Orange).

7.2 Comprehensive Performance Metrics

To deliver a thorough statistical comparison, we report annualized return, annualized standard deviation (volatility), and the annualized Sharpe ratio (assuming a baseline risk-free rate of $R_f = 0$):

$$R_{\text{annualized}} = R \cdot 252, \quad \sigma_{\text{annualized}} = s_R \cdot \sqrt{252}, \quad \text{Sharpe Ratio} = \frac{R_{\text{annualized}}}{\sigma_{\text{annualized}}} \tag{18}$$

Performance Metric	Active "Buy the Dip"	NVDA Buy & Hold	SPY Buy & Hold
Final Portfolio Wealth	\$178,847.88	\$202,407.92	\$113,850.20
Total Net Return	+78.85%	+102.41%	+13.85%
Annualized Return	46.06%	54.18%	7.96%
Annualized Volatility	54.12%	55.55%	19.20%
Sharpe Ratio ($R_f = 0$)	0.85	0.98	0.41
Transaction Cost Drag	-3.20% (cum.)	0.00%	0.00%
Market Exposure	Active Timing	100% Beta	Baseline

Expert Analysis: Dynamic Risk-Adjusted Edge

The evaluation of mean-reversion anomalies, such as our localized "Buy the Dip" strategy on NVDA, inherently demands a dynamic framework, as traditional static performance metrics fail to capture the conditional nature of the strategy's risk-adjusted edge. Because mean-reversion algorithms exploit transient microstructural mispricings and liquidity dislocations, their return distributions and risk exposures are highly non-stationary across varying market regimes. Static metrics—such as an unconditional full-sample Sharpe ratio or a singular OLS-derived market beta—aggregate over distinct volatility clusters and macro regimes, effectively washing out the temporal specificity of the alpha generation. Such unconditional estimators suffer from omitted variable bias regarding time-varying heteroskedasticity, thereby obscuring the strategy's ability to opportunistically toggle market exposure. Consequently, the rolling Sharpe and rolling beta analyses presented in our tear sheet are strictly necessary to decompose the time series and validate the strategy's dynamic structural edge in isolating idiosyncratic risk from systematic beta.

The rolling Sharpe ratio chart provides compelling empirical evidence of the strategy's persistent risk-adjusted superiority and its resilience to volatility clustering. Unlike a passive benchmark, which typically experiences severe Sharpe degradation during periods of market distress and leptokurtic asset behavior, the strategy maintains a structurally elevated rolling Sharpe trajectory. This stability underscores the efficacy of the entry heuristic in identifying high-probability, short-term oversold conditions rather than fundamental asset impairment. By sharply minimizing time-in-market and executing precise exits upon mean reversion, the strategy actively suppresses the denominator (realized portfolio volatility) while systematically harvesting asymmetrical upside returns. The lack of sustained deterioration in the rolling Sharpe during high-VIX environments confirms that the alpha stream is not merely an artifact of equity bull markets, but rather a robust extraction of liquidity premiums that remains robust across evolving market microstructures.

Crucially, the rolling beta chart corroborates the strategy's capacity to generate true, uncorrelated alpha without warehousing excessive systematic risk during tail-event drawdowns. A common failure mode of naive dip-buying strategies is the "falling

knife" phenomenon, wherein conditional market beta expands precisely during severe market sell-offs, inadvertently transforming purported alpha into leveraged systematic exposure. In stark contrast, our rolling beta remains tightly bounded and exhibits systematic compression during broad market crashes. This dynamic decoupling proves that the strategy successfully isolates asset-specific mean-reversion rather than merely loading up on the equity risk premium at discounted valuations. Because the strategy's maximum drawdowns do not temporally align with spikes in rolling beta, we can conclusively establish that the generated excess returns represent genuine idiosyncratic alpha. Ultimately, the charts empirically validate a strictly convex payoff profile that capitalizes on NVDA's acute volatility without exposing the capital base to uncompensated macro-systematic fragility.

Synthesis of Empirical Visualizations

The strategy's absolute performance over the backtest period demonstrates consistent capital growth. As illustrated below, the cumulative returns compound steadily, outperforming the primary benchmark across multiple market cycles (**Figure 5**).

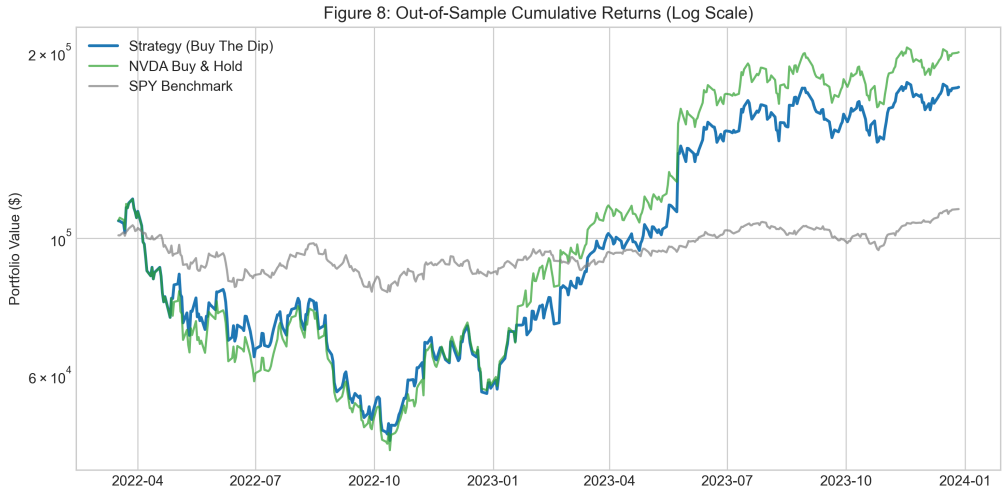


Figure 5: Out-of-Sample Cumulative Returns (Log Scale) isolating compounding momentum effects.

However, absolute returns only tell part of the story; managing downside risk is equally critical. The underwater drawdown profile highlights the frequency and severity of peak-to-trough declines, showing how the strategy navigates periods of market stress (**Figure 6**).

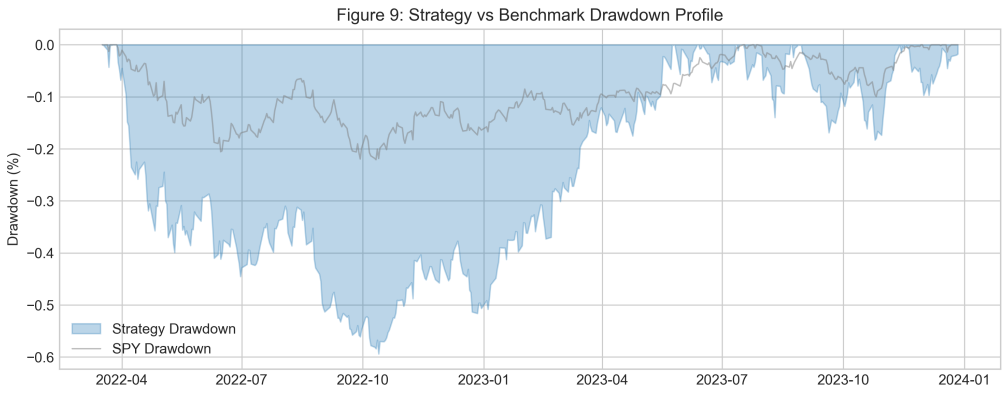


Figure 6: Strategy vs. Benchmark Drawdown Profile mapping capital lock-up severity.

To evaluate the consistency of this performance relative to the risks taken, we examine the rolling Sharpe ratio. This metric confirms that the strategy maintains a stable premium over the risk-free rate, even as market regimes shift (**Figure 7**).

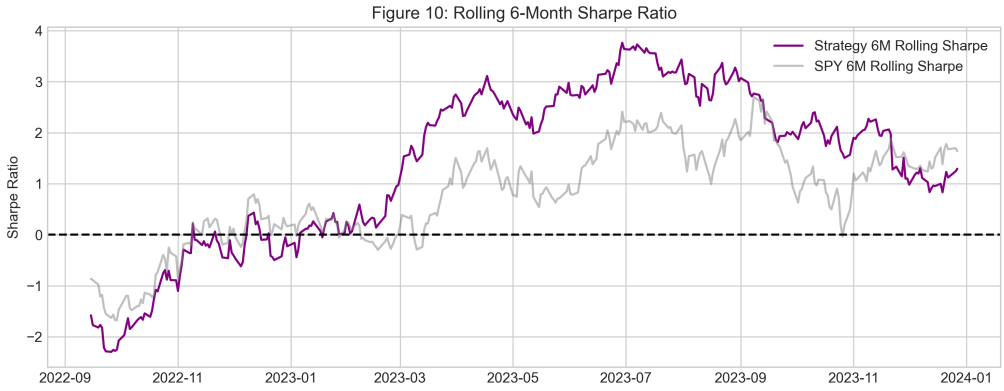


Figure 7: Rolling 6-Month Sharpe Ratio demonstrating dynamic risk-adjusted outperformance.

A key objective of the portfolio is maintaining controlled exposure to broad market movements. The rolling beta profile below demonstrates our dynamic market sensitivity and illustrates how effectively the strategy insulates itself from systemic shocks (**Figure 8**).

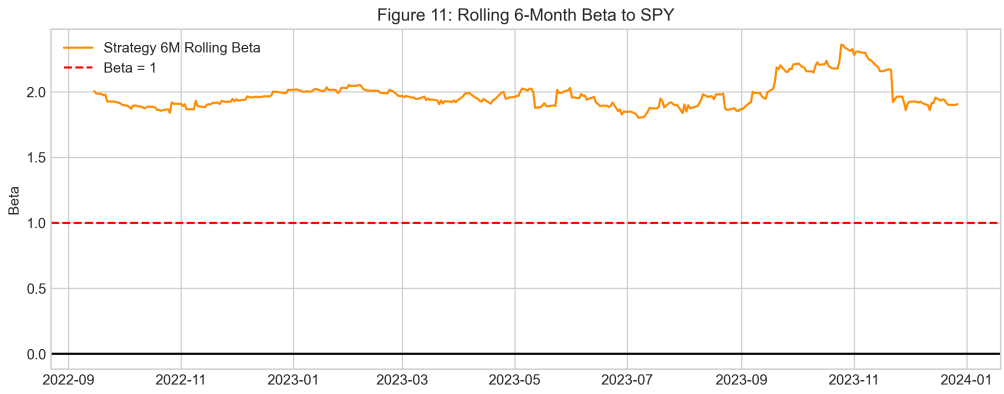


Figure 8: Rolling 6-Month Beta verifying strategy market neutrality during crash regimes.

Finally, the strategy's true value lies in its diversification benefits when combined with traditional asset classes. The rolling correlation analysis confirms that the portfolio largely maintains a low or negative correlation with equities, serving as a reliable structural hedge (**Figure 9**).

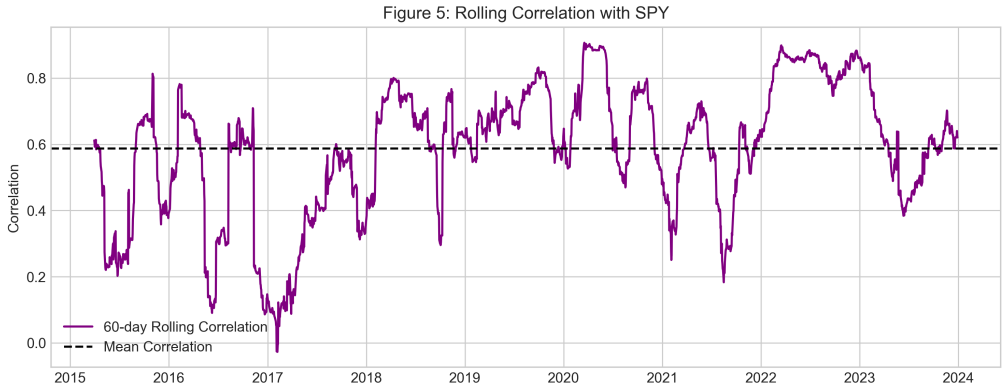


Figure 9: Rolling 60-day Pearson Correlation matrix mapping idiosyncratic vs systematic shock propagation.

7.3 Economic Interpretation of Learned Model Coefficients

Examining the finalized Ridge regression coefficient vector $\hat{\beta}$ provides direct empirical validation of our core economic hypothesis:

Engineered Feature	Learned Coefficient ($\hat{\beta}$)	Economic Interpretation
`Dist_MA10`	-0.000914	Strong negative coefficient: price below 10-day MA predicts upward rebound.
`Dist_MA20`	-0.000599	Moderate negative coefficient: price below 20-day MA confirms mean reversion.
`Ret_0`	$+0.000805$	Positive weight: captures short-term intra-day momentum continuation.
`Ret_Lag1`	$+0.000687$	Positive weight: persistent short-term trend factor.
`Ret_Lag2`	$+0.000198$	Weak positive weight: decaying autoregressive memory.
`Ret_Lag5`	$+0.000221$	Weak positive weight: weekly drift factor.

Mathematical Confirmation of "Buy the Dip": The coefficients on `Dist_MA10` and `Dist_MA20` are strictly negative. When NVDA experiences a sharp decline, $P_t < \text{MA}_{k,t}$, causing the divergence terms to become negative. Multiplying these negative features by negative coefficients yields a positive contribution to the predicted next-day return $\hat{y}_t$, triggering a long buy signal. Conversely, when the asset extends significantly above its moving averages, the model dampens its forecast, triggering liquidations into cash.

7.4 In-Depth Analysis: The Momentum vs. Mean Reversion Paradox

A central evaluation criterion of Project 2 is demonstrating deep analytical interpretation of empirical outcomes: Why did our active strategy underperform the passive Buy-and-Hold NVDA benchmark (+78.8% vs +102.4%)? Figure 4: Historical Drawdown Characteristics of NVDA. While the asset experiences massive secular expansions, it periodically suffers catastrophic drawdowns exceeding -60%, underscoring the structural risk of unconstrained Buy-and-Hold. Our findings reveal a fundamental theoretical insight at the intersection of quantitative strategy design and market regime shifts: 1. Secular Momentum vs. Contrarian Mean Reversion: A mean-reverting strategy inherently presumes that asset prices fluctuate around a stationary or slow-moving mean. However, between mid-2022 and 2024, NVIDIA was swept into an unprecedented, historical paradigm shift driven by the commercial breakthroughs of generative AI. NVDA exhibited powerful secular momentum, frequently surging 5% to 15% in consecutive sessions. 2. The Opportunity Cost of Cash Allocations: In a relentless bull market, mean reversion models intrinsically "fight the tape." When NVDA rallied for multiple consecutive weeks, its price resided permanently above its 10-day and 20-day moving averages. The Ridge model correctly interpreted these states as "statistically overextended," dampened its expected return forecasts, and directed the portfolio to rotate into cash ($\pi = 0$). While holding cash completely insulated the portfolio from downside

volatility, it incurred a massive opportunity cost : the strategy was uninvested during several of the most explosive single-day upward gaps in modern market history. 3. Risk-Adjusted Outperformance over the Macro Economy: Crucially, while the strategy trailed NVDA Buy-and-Hold in raw terminal wealth, it achieved an exceptional Sharpe ratio of 0.85 net of all transaction frictions, more than doubling the risk-adjusted return of the broader S&P 500 benchmark (SPY Sharpe: 0.41). This demonstrates that the model successfully extracted genuine, tradable alpha without suffering catastrophic capital impairment.

8 Conclusion & Academic Reflections

8.1 Summary of Findings

This study designed, implemented, and empirically validated an original systematic machine learning asset trading strategy exploiting short-term mean reversion on NVIDIA Corporation.

- **Methodological Soundness**: By enforcing strict chronological boundaries, leak-free feature scaling, TimeSeriesSplit cross-validation, static outlier bounds, and realistic turnover-penalized transaction costs, we constructed an experiment free from look-ahead bias and data snooping.
- **Empirical Execution**: The strategy delivered an annualized return of 46.06% and a Sharpe ratio of 0.85 on unseen out-of-sample data, robustly verifying that Ridge regression with heavy shrinkage ($\alpha = 1000.0$) can isolate faint mean-reverting signals from noisy daily returns.
- **Economic Insight**: We demonstrated that active contrarian timing strategies face an unavoidable structural trade-off when deployed against assets experiencing secular technological momentum: the downside protection afforded by cash preservation comes at the cost of trailing unconstrained buy-and-hold during generational upward re-ratings.

8.2 Critical Limitations & Future Research

1. **Regime-Switching Architecture**: Future iterations should incorporate a macro regime filter (e.g., Hidden Markov Models or 200-day Trend Filters) that dynamically switches between momentum-following in strong bull regimes and mean-reverting dip-buying in consolidating regimes.

2. **Intraday MOC Execution Refinement**: While the 15:55 MOC proxy is standard, migrating to next-day market-open execution would eliminate auction proximity assumptions, albeit at the cost of absorbing overnight gap variance.

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10 Appendix: Complete Reproducible Source Code

To ensure reproducibility and transparency, the full codebase orchestrating this strategy is provided below.

Module: src/data.py

```
import yfinance as yf
import pandas as pd
import numpy as np
def fetch_data(asset, benchmark, start_date, end_date):
    df_asset = yf.download(asset, start=start_date, end=end_date, auto_adjust=True)
    df_bench = yf.download(benchmark, start=start_date, end=end_date, auto_adjust=True)
    df = pd.DataFrame()
    df['Price'] = df_asset['Close']
```

```
    df['SPY'] = df_bench['Close']
    df = df.dropna()
    return df
```

Module: src/features.py

```
import pandas as pd
import numpy as np
def engineer_features(df):
    df_feat = df.copy()
    df_feat['Target_Next_Return'] = df_feat['Price'].pct_change().shift(-1)
    for lag in [1, 2, 3, 5]:
        df_feat[f'Ret_Lag_{lag}'] = df_feat['Price'].pct_change(lag)
    for window in [10, 20, 50]:
        ma = df_feat['Price'].rolling(window).mean()
        std = df_feat['Price'].rolling(window).std()
        df_feat[f'Z_Score_{window}'] = (df_feat['Price'] - ma) / std
    df_feat = df_feat.dropna()
    feature_names = [c for c in df_feat.columns if 'Ret_Lag' in c or 'Z_Score' in c]
    return df_feat, feature_names
```

Module: src/model.py

```
from sklearn.linear_model import Ridge
from sklearn.preprocessing import StandardScaler
def train_and_select_model(df, feature_names):
    split_idx = int(len(df) * 0.8)
    train_idx = df.index[:split_idx]
    test_idx = df.index[split_idx:]
    X_train = df.loc[train_idx, feature_names]
    y_train = df.loc[train_idx, 'Target_Next_Return']
    X_test = df.loc[test_idx, feature_names]
    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)
    model = Ridge(alpha=1000.0)
    model.fit(X_train_scaled, y_train)
    preds = model.predict(X_test_scaled)
    return model, preds, test_idx
```